Screen2Sleep

Data Preparation/Feature Engineering

1. Overview

Screen2Sleep AI is an Android assistant that warns a user, late in the evening, how much longer the app session they are in is likely to last and what that may cost in sleep opportunity. **Model A** answers the first half of that question (research question RQ1): given an app session that has already lasted **$t \geq 5$ minutes**, predict its **remaining time R** as a median together with a 25–75% range.

The raw data for Model A are not sessions but a stream of app events (opened, closed, interaction). The data preparation phase therefore does most of the modeling work: it decides what a "session" is, how long it lasted, and whether its end was actually observed. Three properties of the problem drive every decision below:

- **Censoring.** Some sessions have no observed end. Dropping them would bias durations downwards, so they are kept as right-censored observations and modeled with survival methods.
- **No look-ahead.** Every feature must be computable on the phone at the prediction moment (session start + t). History features use only the user's earlier sessions.
- **On-device deployment.** Whatever is learned has to ship to Android as a small JSON parameter file and be reproduced exactly in Kotlin, which favours simple, closed-form transformations.

The second component, Model B (RQ2), examines whether a person's evening phone use above their own usual level is associated with a difference in subsequent sleep outcomes. The motivation is supported by research reporting associations between bedtime mobile-phone use and sleep measures, although an observational dataset cannot establish that phone use itself causes a later sleep onset (Exelmans & Van den Bulck, 2016). Model A estimates the amount of an ongoing app session that remains; Model B is intended to translate an excess-use estimate into a person-specific, appropriately qualified sleep-related estimate. The two components use separate datasets and participants, and their outputs should only be linked through a clearly defined measure of additional phone use rather than by treating records across the datasets as matched observations.

This design requires two complementary forms of preprocessing. LSApp provides second-level foreground-app event logs suited to reconstructing session durations and modelling right-censored remaining time. GLOBEM provides longitudinal, participant-day features derived from phone and wearable sensing, allowing within-person comparisons of evening screen-unlock duration and subsequent sleep measurements (Aliannejadi et al., 2021; Xu et al., 2022, 2023). The available GLOBEM files currently cover the 2018 cohort only; the full resource spans four annual cohorts from 2018 to 2021. Therefore, analyses of the supplied data must not be presented as estimates from all four years.

The main challenge for Model B is not only missing data but also measurement validity. The released phone features measure screen-unlock episodes, which are a proxy for phone use rather than a direct measure of social-media activity or phone use in bed. Likewise, Fitbit-derived sleep-onset fields are not equivalent to a clinical sleep-onset measure and may contain device-dependent artefacts. Validation studies show that wearable sleep trackers can be less reliable at distinguishing wakefulness from sleep, so candidate outcomes are screened before modelling (Chinoy et al., 2021). The final wording of the

app's warning will depend on which outcome can be measured reliably. Any estimated relationship from GLOBEM will be described as an association, not a demonstrated causal effect.

The Model A pipeline is implemented in ml/model_a/ (sessions.py → features.py → train.py, with evaluate.py and personal.py), and its report figures are generated by python -m ml.model_a.report_figures. The Model B pipeline is implemented in ml/model_b/globem_model_b.py; it runs only on the credentialed team member's computer and writes aggregate tables, figures and parameters. Its target choice is reported in section 7.2 and its held-out results in Model Exploration section 5.3. The objective of this phase is to produce reproducible data definitions, leakage-safe features and evidence-based model comparisons for both components.

The intended Model B outcome was initially sleep-onset latency. However, an initial inspection of the supplied 2018 GLOBEM cohort identified 7,443 zero observations among 7,476 nonmissing values in the candidate sleep-onset field (99.6%). This variable therefore fails the previously specified target-quality rule and cannot support the intended numerical sleep-onset-delay prediction. The present analysis instead evaluates Fitbit main-sleep duration as a distinct exploratory outcome. This change must not be interpreted as evidence that phone use causes shorter sleep. In addition, Model A predicts the remainder of one foreground-app session, while Model B uses total screen-unlock duration during an entire evening; these measurements are not interchangeable without a separately validated exposure-alignment step.

2. Data Collection

2.1 LSApp: App-session prediction (Model A)

Model A is trained on the public **LSApp** dataset (Aliannejadi et al., 2021, ACM TOIS 39(3); github.com/aliannejadi/LSApp, file lsapp.tsv.gz). LSApp was collected from Android phones of volunteer participants by a logging app that recorded which app was in the foreground. The dataset has no licence file; its README asks for citation, so it is used for academic purposes only and is not redistributed.

Table 1. *LSApp dataset facts.*

Item	Value
Size	3,658,589 rows, 5 columns, tab-separated UTF-8, no missing values
Columns	user_id, session_id, timestamp, app_name, event_type
event_type	Opened 45.7%, Closed 45.6%, User Interaction 8.6%, Broken 0.1%
Users and period	292 users, 2017-09-09 to 2018-05-17; median 5 days per user
Apps	87 named apps, no category column
Timestamp	YYYY-MM-DD HH:MM:SS, 1 s resolution, no time zone

Two properties of the raw file matter for preprocessing. First, session_id is a **phone** session (screen on to screen off), not an app session: 60.7% of phone sessions contain more than one app, so app sessions have to be reconstructed from the events. Second, the file is distributed as a gzip-compressed tar archive; load_events() reads it directly, keeps the five columns and parses timestamps with an explicit format.

Apps carry no category, so we added a manual mapping (shared/app_categories.json) of the 87 app names into 10 categories: social, messaging, calls, browser/search, email, media, rewards, games,

shopping/finance and a default "utility" category. The same file will be extended with Android package names so that the phone uses identical categories.

2.2 GLOBEM: Longitudinal phone and sleep data (Model B)

Model B uses GLOBEM (GLOBEM Dataset: Multi-Year Datasets for Longitudinal Human Behavior Modeling Generalization), released through PhysioNet and introduced at the NeurIPS 2022 Datasets and Benchmarks Track (Xu et al., 2022, 2023). The underlying longitudinal study involved undergraduate participants at the University of Washington across four annual spring cohorts: INS-W_1 (2018, 155 person-years), INS-W_2 (2019, 218), INS-W_3 (2020, 137) and INS-W_4 (2021, 195). These add to 705 participant-year records from 497 unique people; the annual totals must not be mistaken for 705 distinct individuals. The study followed participants for approximately ten weeks per year and collected passive sensing data from an AWARE-based phone application and Fitbit devices alongside survey responses.

Phone sensing included screen-status/unlock episodes, location, call records and Bluetooth activity, while Fitbit supplied physical-activity and sleep features. The original GLOBEM resource also provides pre/post-study surveys and ecological momentary assessment (EMA) responses. For this project, the primary analytical inputs are the phone-usage and sleep feature families, with steps and other contextual streams available for exploratory checks or sensitivity analyses; including further variables in a predictive model requires a separate relevance, missingness and leakage assessment. Importantly, we are using precomputed feature-level CSVs rather than the identifiable, raw sensor streams. These features were derived using the Reproducible Analysis Pipeline for Data Streams (RAPIDS), whose published methodology supports consistent feature extraction across mobile and wearable sources (Vega et al., 2021).

GLOBEM provides repeated participant-day summaries rather than an event-by-event app-session log. Each sensor-feature file is indexed by an anonymised participant identifier (pid) and date. Features are supplied for morning (06:00–12:00), afternoon (12:00–18:00), evening (18:00–24:00), night (00:00–06:00), whole-day and rolling-history windows. Multiple feature versions are available; for this report the unnormalised minute-valued features are preferable wherever direct interpretation and future Android implementation are required. The evening exposure of interest is the duration of screen-unlock episodes, not an observation of the exact app being used, and it does not establish whether use occurred after the person went to bed.

2.3 Uploaded GLOBEM subset and verified structure

An initial structural audit of the uploaded files identified the 2018 INS-W_1 cohort. All seven supplied daily sensor-feature CSVs contain 14,260 rows, 155 unique pid values and 92 distinct stored dates (03 April–03 July 2018), with no duplicate (pid, date) keys. The date range is the range encoded in the supplied files, not a literal reconstruction of the participants' actual calendar dates: the GLOBEM creators intentionally shifted dates by whole weeks for privacy, preserving temporal order and day of week within an individual (Xu et al., 2023).

Uploaded subset audit — counts below were computed from the supplied files.

File	Rows	Columns	Role in project
screen.csv	14,260	1,137	Evening screen-unlock duration and frequency

sleep.csv	14,260	921	Fitbit main-sleep outcomes and bedtime
steps.csv	14,260	597	Potential physical-activity covariates
location.csv	14,260	1,110	Potential context and sensitivity analyses
call.csv	14,260	786	Potential contextual features
bluetooth.csv	14,260	894	Potential contextual features
wifi.csv	14,260	84	Contextual features; evaluate missingness
platform(1).csv	201	2	Participant platform metadata; subset before join

The platform metadata contain 201 participants, of whom all 155 participants in the daily 2018 files can be matched. After restricting platform.csv to these 155 participants, the distribution is 145 iOS and 10 Android users. The extra 46 platform entries must not be counted as participants in the analysed 2018 daily panel. The platform imbalance is relevant to the transfer of a Model B relationship to an Android-only product.

2.4 Initial variable-level audit and preprocessing decisions

The initial read stage checks the CSV encodings, preserves the original feature names, parses pid/date, removes the non-analytical exported index column, confirms unique keys and limits the participant-platform merge to identifiers present in the selected cohort. Because GLOBEM already contains extracted features, data collection preprocessing is primarily a schema and alignment task; the rules for missing values, implausible values and train/test-dependent transformations are discussed under Data Cleaning and Data Transformation.

For the main exposure, f_screen:phone_screen_rapids_sumdurationunlock:evening records the total minutes of screen-unlock episodes between 18:00 and 24:00. In the supplied cohort this field has 9,960 nonmissing values (69.8% of participant-days), with a median of approximately 86.4 minutes among observed values. The matching unlock-count feature has the same number of observed rows. These values describe the subset available for analysis rather than the entire four-year GLOBEM resource.

The planned main-sleep sleep-onset field, f_slp:fitbit_sleep_summary_rapids_sumdurationtofallasleepmain:allday, has 7,476 nonmissing values (52.4% of participant-days). However, 7,443 of those observed values (99.6%) are zero. This is too little measured variation to treat the field as a reliable continuous response without further validation. Alternative outcomes already present in the files include total main-sleep duration (median 429 minutes among 7,476 observed values) and time awake within the main sleep episode (median 23 minutes); the latter reflects wakefulness during sleep and must not be relabelled as sleep-onset latency. On identical stored (pid, date) values, evening unlock duration and main-sleep duration have 6,324 candidate paired records before any overnight-date-alignment, minimum-history or plausibility filtering. These are not yet final training observations.

RAPIDS assigns Fitbit sleep episodes to daily summaries using the configured "last night end" convention. The firstbedtime feature is expressed in minutes relative to the reference midnight, and valid overnight bedtimes may exceed 1,440 minutes rather than indicating invalid data (RAPIDS, n.d.-a). A same-pid, same-date join was used to construct candidate pairs; 6,324 records have both evening screen-unlock duration and main-sleep duration. To reduce temporal ambiguity in a conservative primary analysis, we additionally restrict the sample to records with a recorded main-sleep bedtime at or after 1,440 minutes, so that the full 18:00–24:00 exposure window precedes the recorded bedtime. This restriction selects later-bedtime participants and is therefore treated as a design choice to be checked in sensitivity analyses, not a universally unbiased definition of night-level exposure. The study's stored dates are privacy-shifted and are used only for within-person order and day-of-week comparisons.

2.5 Access, confidentiality and scope

GLOBEM is a credentialed PhysioNet resource governed by the PhysioNet Credentialed Health Data License, version 1.5.0. It prohibits reidentification and sharing access to restricted participant data, requires reasonable security safeguards and limits use to lawful scientific research. Processing should be performed only by authorised researchers under approved arrangements, with no release of row-level participant data or files in the public project repository. The exact source release version for the uploaded copies should be verified against the original download record before the final dataset citation is fixed; the official PhysioNet page currently documents version 1.1 (Xu et al., 2023). Results published in this report should use aggregate statistics only.

Finally, the dataset is observational and its participants are predominantly college students, while the files currently supplied represent only one cohort. The model should therefore be evaluated for missingness, participant-level imbalance, temporal dependence and transferability before the results are generalised. In particular, any association between evening screen-unlock time and sleep does not demonstrate a causal sleep delay, and quantitative claims about Model B must await a validated outcome and held-out evaluation.

3. Data Cleaning

3.1 LSApp Data Cleaning

The dataset has no missing values, so cleaning is about event-stream quality: duplicate rows, unmatched open/close events, fragmented sessions and time zone. Five rules were fixed as project decisions and are reused unchanged on Android:

1. **Remove only consecutive exact duplicates.** A row is dropped only if it is identical to the row directly before it (24.4% of rows; 2,765,688 remain). Removing all duplicate rows would delete 59.7% of rows, including legitimate repeated events, and artificially lengthen sessions.
2. **Pair events per (session_id, app_name).** An Opened with no open session starts one, an Opened while one is open is ignored, and the next Closed ends it. This yields 1,160,811 raw sessions, of which 98.3% receive a Closed event.
3. **Keep unmatched sessions as right-censored.** A session with no Closed event is not dropped; it ends at the last time the app is known to be in the foreground and is flagged event = 0. Broken events are not treated as automatically censored.

4. **Merge re-entries within 30 s.** Sessions of the same user and app that restart within 30 seconds of the previous end are merged into one (for example, briefly switching to a notification and coming back).
5. **Time zone.** LSApp timestamps carry no documented time zone. We treat them as UTC and shift them by −5 h for time-of-day features; this is a modelling convention, not a verified property of the collection. An unshifted rerun (--utc-offset 0) leaves the t = 5 results practically unchanged (MAE 3.18 vs 3.18 min, chronological) and only changes which sessions fall into the evening subgroup. On the phone the device's own local time is used.

Result: 221,843 app sessions, median duration 26 s, 7.2% zero-second sessions and 5.9% censored. Zero-second sessions are kept because they are real events (an app opened and immediately left); they never enter the training set, because Model A only predicts for sessions that have reached 5 minutes. 595 censored sessions are known to have lasted more than 5 minutes; dropping them, as an earlier analysis did (213,028 sessions), would remove exactly the long sessions the nudge is meant for.

Outliers. Very long sessions (up to several hours) were not clipped. They are rare, partly censored and exactly what the warning targets; the log-scale target and the survival likelihood keep them from dominating the fit. They are instead reported separately (landmarks t = 15 and 30 min).

The 30-second merge gap is the single most influential cleaning parameter. With a gap of 0 s only 17 sessions exceed 30 minutes, against 140 at 30 s; long sessions only appear once quick re-entries are joined (Figure 2). 30 s was chosen as a conservative value: it joins brief interruptions such as checking a notification, while longer gaps increasingly join what are plausibly separate visits and inflate the long-session counts further (Table 2).

Table 2. *Sensitivity of session counts to the re-entry merge gap.*

Merge gap	Sessions	> 5 min	> 15 min	> 30 min
0 s	337,338	5,383	227	17
5 s	264,833	7,130	538	110
10 s	249,871	7,692	578	117
30 s (chosen)	**221,843**	**9,337**	**738**	**140**
60 s	201,170	11,436	929	164
120 s	179,282	15,164	1,481	227

3.2 GLOBEM Data Cleaning (2018 Cohort)

The GLOBEM CSV files already contain RAPIDS-engineered features. Unlike LSApp, their primary quality concern is missing sensor observations rather than unmatched foreground-app events. The seven sensor-feature tables contain one row per (pid, date), with no duplicate keys in the supplied 2018 extract. Data processing retains the original unnormalised evening-unlock and main-sleep variables, parses dates, removes the exported index column, and joins measurements only by their anonymised participant and stored date identifiers. Missing target or exposure values are not imputed.

A preliminary eligibility audit starts with 14,260 participant-day rows (155 participants). Evening-unlock duration is observed for 9,960 rows and main-sleep duration for 7,476. The intersection is 6,324 candidate pairs. Keeping evening use between 0 and 360 minutes and main-sleep duration

between 180 and 840 minutes leaves 6,122 pairs. Requiring recorded main-sleep bedtime of at least 1,440 minutes reduces the set to 4,759 pairs from 136 participants. Requiring at least seven eligible recorded nights per participant leaves 4,718 pairs from 124 participants. All values represent the uploaded 2018 cohort only; they are not totals for the four-cohort GLOBEM resource.

The post-midnight restriction has a clear limitation: it prioritises temporal order at the cost of excluding earlier-bedtime participants. Missingness can also depend on device adherence and sleep routines. We therefore distinguish this analysis subset from a population-representative sample and do not treat absent observations as zero minutes. Eligibility thresholds are applied before the chronological split to define a stable study subset, whereas personal means, imputations and all other learned prediction features are fitted only from the training portion. Future sensitivity analyses should compare alternative bedtime windows and missingness patterns without using outcomes that occur after the prediction moment.

Table 2b. GLOBEM sample attrition (2018 cohort)

Filter	Observations	Participants (where assessed)
Participant-day grid	14,260	155
Screen + sleep both observed	6,324	136
Exposure 0–360 min; sleep 180–840 min	6,122	136
Main bedtime ≥ 1,440 min	4,759	136
At least seven eligible records per participant	4,718	124

4. Exploratory Data Analysis (EDA)

Session durations are extremely skewed. The median app session lasts 26 s and only 4.2% of sessions (9,337) reach the 5-minute landmark at which Model A starts predicting (Figure 1). This motivates both the log transformations and the decision to model only sessions that are already long.

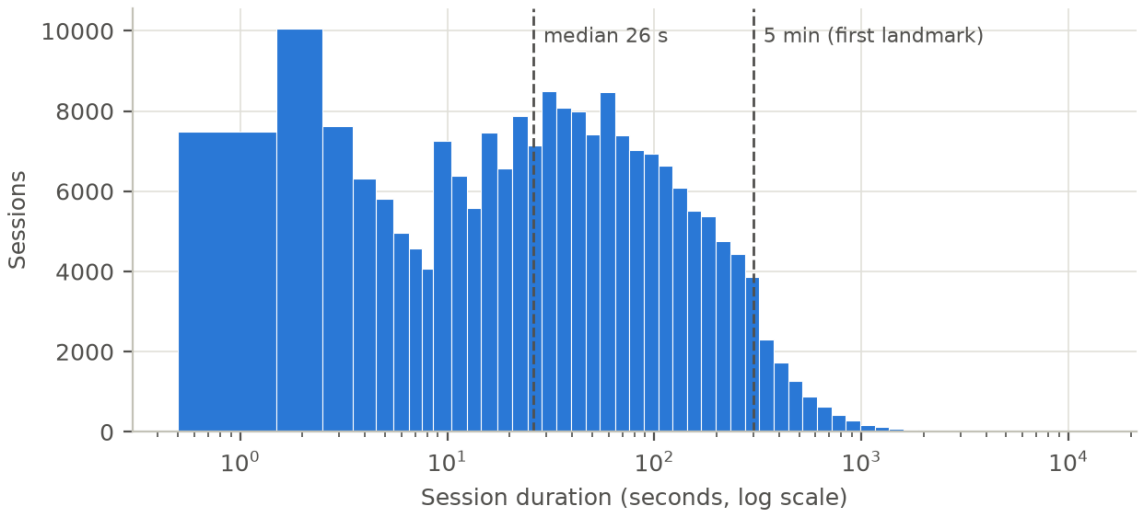


Figure 1. *Distribution of app session durations after reconstruction (log scale; zero-second sessions excluded).*

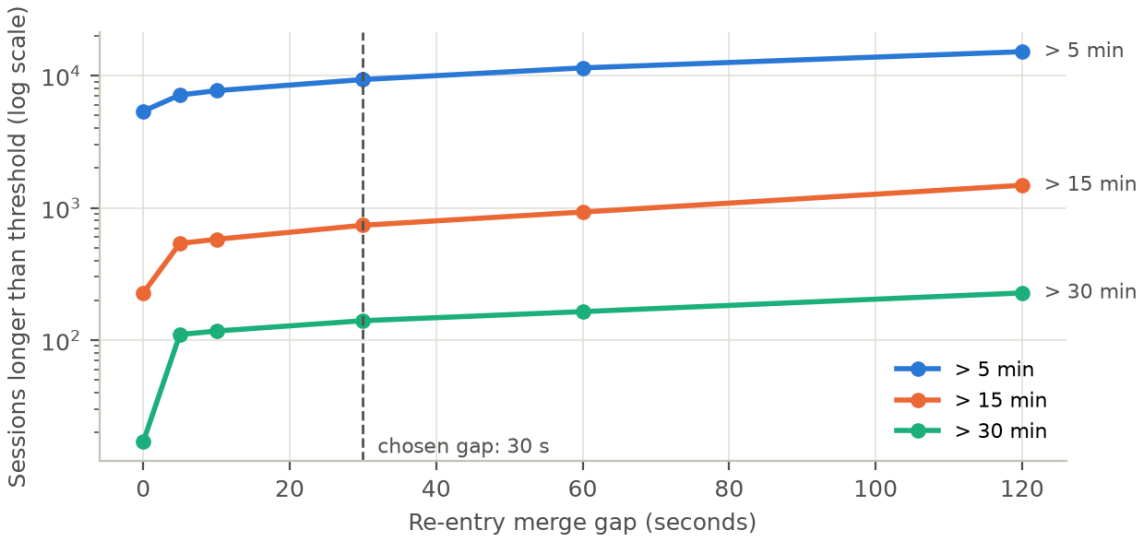


Figure 2. *Number of sessions longer than 5, 15 and 30 minutes as a function of the re-entry merge gap.*

The remaining time grows with elapsed time. Among sessions that reached 5 minutes, the Kaplan-Meier median remaining time is 2.1 min; among those that reached 30 minutes it is 10.4 min (Table 3, Figure 3). The longer a session has lasted, the longer it is likely to continue, so elapsed time is a key feature and the hazard of ending is decreasing over time. This matches a Weibull shape parameter below 1.

Table 3. *Remaining time by elapsed landmark (Kaplan-Meier estimates, censoring handled).*

Reached	Sessions	Median remaining (min)	Q1–Q3 (min)
5 min	9,337	2.1	0.8–4.7
15 min	738	5.6	2.0–12.3
30 min	140	10.4	4.0–28.7

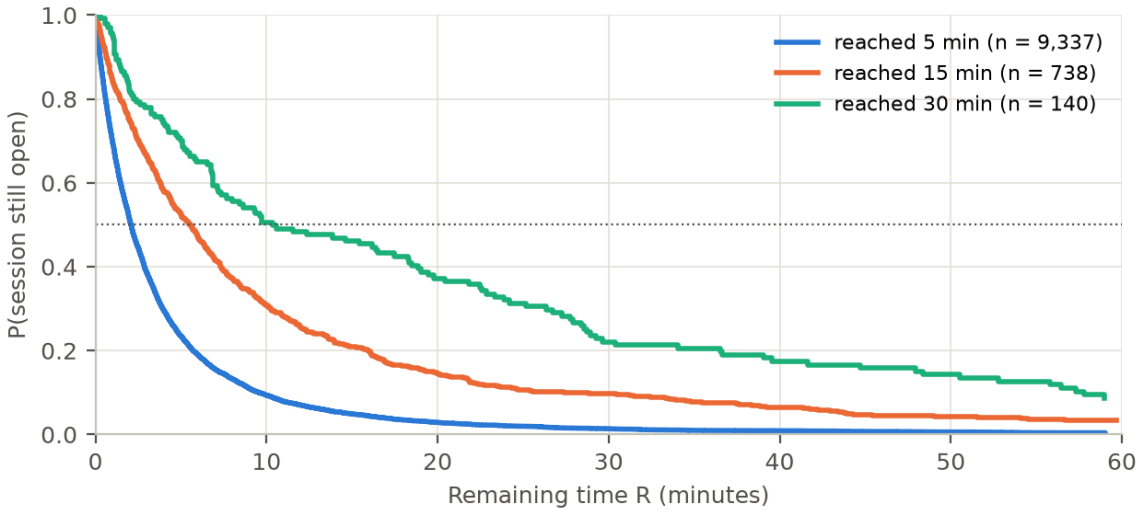


Figure 3. *Kaplan-Meier estimates of remaining time for sessions that reached 5, 15 and 30 minutes (censoring handled).*

App category matters. After 5 minutes, the Kaplan-Meier median remaining time ranges from 1.0 min for email to 4.25 min for games, with rewards (3.05) and social (2.5) also above average (Figure 4). Social apps contribute 43% of the 5-minute snapshots.

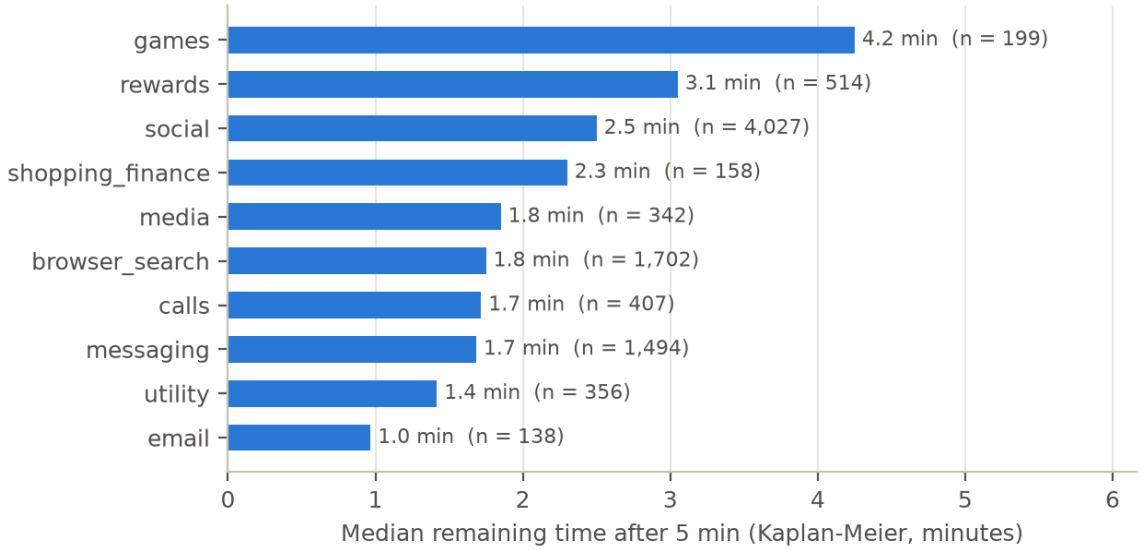


Figure 4. *Median remaining time after 5 minutes by app category (Kaplan-Meier), with the number of sessions.*

Evenings produce more long sessions. Between 20:00 and 01:00 local time, 4.8–5.4% of sessions pass 5 minutes, against 3.6–4.0% during the working day (Figure 5). The evening window is the use case of the product, so it is evaluated separately.

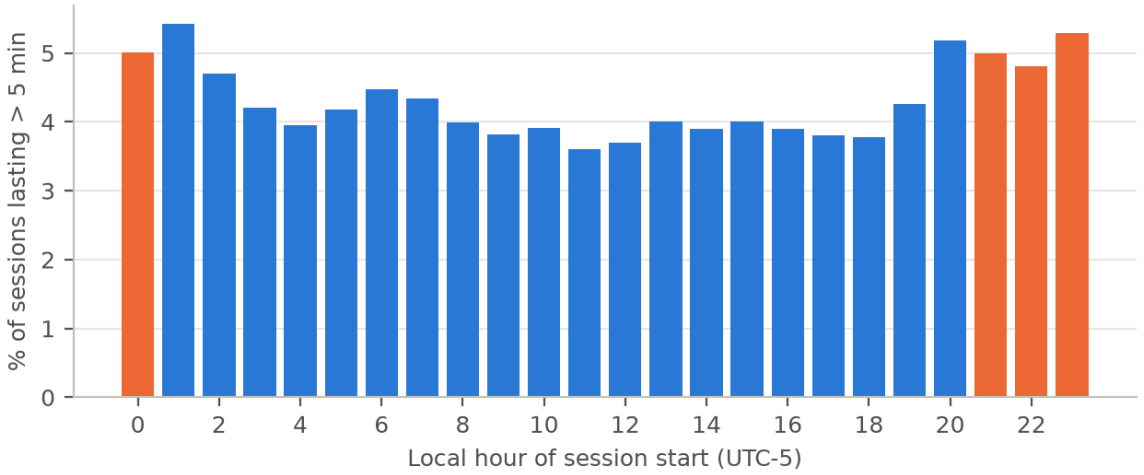


Figure 5. *Share of sessions that last longer than 5 minutes by local hour of session start; the evening window (21:00–01:00) is highlighted.*

Known data limitations. Only 140 sessions reach 30 minutes, and 38.6% of sessions longer than 30 minutes come from a single reward app (Swagbucks), whose usage pattern (watching videos for points) differs from ordinary leisure use. We therefore report a sensitivity analysis without reward apps (Model Exploration, section 3). LSApp contains no TikTok, and timestamps have no time zone.

5. Feature Engineering

Landmark design. Each session that is still open at $t \in \{5, 10, 15, 20, 30\}$ minutes contributes one snapshot row per landmark, with target $R = \text{duration} - t$ and the censoring flag carried over. This gives 12,566 snapshots (9,337 / 1,981 / 738 / 370 / 140 per landmark; 6.3% censored). A single model then covers every elapsed time, and the phone can re-predict as the session continues. Landmarks below 5 minutes are never used, because no warning is shown there. Every feature uses only information available at the prediction moment start + t:

Table 4. *Model A features.*

Feature	Definition	Rationale
log_elapsed_min	ln(t)	Remaining time grows with elapsed time (Figure 3).
hour_sin, hour_cos	sin/cos(2π · local hour / 24) at start + t	Cyclic encoding so 23:59 and 00:00 are close; evening effect (Figure 5).
is_weekend	1 on Saturday or Sunday	Weekend leisure use is less constrained.
log_use_today_min	ln(1 + minutes of earlier sessions today, counted only up to this session's start)	Captures binge days and fatigue.
log_gap_prev_min	ln(1 + minutes since previous session ended)	Rapid app switching versus a deliberate new session.
hist_log_n	ln(1 + number of earlier sessions)	How much personal history backs the history features.
hist_app_mean_logD	Mean ln(1 + duration_s) of the user's earlier completed sessions of this app	Personal habit for this specific app.
hist_lm_logR	Mean ln(R) of the user's earlier completed sessions that also reached t and had ended before start + t	Personal version of the second baseline.
cat_* (9 dummies)	One-hot app category, reference "utility"	Category differences (Figure 4).

Leakage control. Ordering sessions by start time alone does not guarantee that their outcome is known at the prediction moment, because sessions of different apps can overlap (11.3% of sessions start while an earlier one is still open, mostly through re-entry merging and censored ends). The history features therefore follow three rules: (1) hist_lm_logR uses only earlier sessions that had **ended before start + t**, ordered by their end time; (2) "use so far today" counts earlier sessions only up to the current session's start, so a session that is still open contributes only its minutes so far; (3) duration-based history (hist_app_mean_logD, hist_lm_logR) uses **completed sessions only**, because a censored duration is a lower bound rather than the true length. Same-app sessions never overlap after the 30 s merge, so earlier sessions of the same app have always ended. An audit of an earlier version that ordered by start time found 44 snapshots (0.35%) whose hist_lm_logR used a session still running, and 630 snapshots (5.0%) whose daily use counted a median 0.4 minutes after the session start; both are fixed, and the rules are asserted by automated tests (ml/tests/test_features.py). The three history features that can be undefined for a new user (log_gap_prev_min, hist_app_mean_logD, hist_lm_logR) are filled as described in section 6.

Features considered and not used. Raw app identity (87 levels) was replaced by the 10 categories to avoid sparse levels and to transfer to apps that are not in LSApp. Sequence models over the event stream (LSTM/GRU) are left as future work, because they cannot be reproduced cheaply on the phone.

6. Data Transformation

- **Log transforms.** Durations, gaps and counts are heavily right-skewed (Figure 1), so they enter as ln(1 + x); elapsed time enters as ln(t). In an accelerated failure time model this makes each coefficient act multiplicatively on time.
- **Cyclic encoding** of the hour of day as sine and cosine.
- **One-hot encoding** of the app category (9 dummies, utility as the reference) and a binary weekend flag. Binary columns are not scaled.
- **Missing-value fill.** History features missing for a user's first sessions are filled with their training-set mean, so a new user starts at the population average.
- **Standardisation.** Numeric features are z-scored with the training-set mean and standard deviation. The fill values, means and standard deviations are fitted on the training split only and are exported in params/model_a.json, so Kotlin applies the identical transformation.

```
# train.py - fitted on the training split only, exported to params/model_a.json
def fit_scaler(tr):
    num, _ = feature_names()
    fill = {c: float(tr[c].mean()) for c in HISTORY}    # new user -> population mean
    filled = tr[num].fillna(fill)
    return {"fill": fill, "mean": filled.mean().to_dict(),
            "std": filled.std().replace(0, 1).to_dict()}

def transform(d, scaler):
    num, binary = feature_names()                    # binary: is_weekend, cat_*
    x = d[num].fillna(scaler["fill"])
    x = (x - pd.Series(scaler["mean"])) / pd.Series(scaler["std"])   # z-score
    return pd.concat([x, d[binary]], axis=1)
```

The target is not transformed explicitly: the Weibull AFT and XGBoost AFT models are fitted on R directly and model ln(R) internally, which also handles censoring correctly.

7. Model B: Exploratory Association Between Evening Screen-Unlock Duration and Main-Sleep Duration (GLOBEM)

Model B originally targeted the association between excess evening smartphone use and sleep-onset latency (RQ2). The available GLOBEM data do not support a reliable numerical sleep-onset response: the proposed main-sleep onset feature is zero in 99.6% of its nonmissing rows. The primary analysis reported here therefore evaluates whether within-person variation in evening screen-unlock duration is associated with the duration of the subsequent main sleep episode. This is a revised, observational research question, not a substitute measurement of sleep-onset delay.

Status. The uploaded GLOBEM 2018 INS-W_1 cohort has undergone a variable-level audit and a preliminary reproducible model evaluation. Results are based on the 2018 files only. They do not represent the complete four-cohort resource, and they do not justify presenting additional phone use as a proven cause of reduced sleep. Any deployment of a numeric Model B nudge remains conditional on stronger predictive validity and congruent mobile-device measurement.

7.1 Data Collection

Model B was examined using the INS-W_1 (2018) cohort of the **GLOBEM** dataset (Xu et al., 2022, 2023; physionet.org/content/globem; 2018 INS-W_1 extract, release version subject to verification). The full resource has four yearly cohorts; the supplied files contain only the 2018 cohort, collected from undergraduate students at a US university with a phone sensing app (AWARE) and a Fitbit wristband, and is distributed as features already aggregated to one row per participant and day by the RAPIDS pipeline.

Table 9. *GLOBEM INS-W_1 (2018) subset used for Model B, as found in the supplied files.*

Item	Value
Participants	155 participants, one person-year each; the other three cohorts of the full resource (497 people, 705 person-years in total) are not used
Period	92 stored dates (03 April–03 July 2018, shifted by whole weeks for privacy); 14,260 participant-days
Devices	AWARE app (145 iOS and 10 Android participants); Fitbit Flex2
Format	CSV files of daily RAPIDS features indexed by pid and date; segments morning (06–12), afternoon (12–18), evening (18–24), night (00–06) and all day
Missing values	Evening screen-unlock duration observed on 69.8% of participant-days; main-sleep summary on 52.4%
Licence	PhysioNet Credentialed Health Data License 1.5.0: academic use only, no redistribution

Data handling. The files are processed only on the computer of the credentialed team member, on an encrypted disk, without cloud storage or AI services. Only aggregate statistics and model parameters leave that computer, and no GLOBEM-derived parameters are shipped in the pilot build.

7.2 Target Definition and Cleaning Decisions

The initial target hierarchy prioritised Fitbit main-sleep sleep-onset latency if fewer than 80% of observed measurements were zero and at least half of participants displayed meaningful variation. The inspection of the supplied file found 7,443 exact zeros in 7,476 nonmissing observations (99.6%), so this target fails the zero-rate criterion decisively. Minutes awake within the sleep episode were considered as a secondary outcome but do not specifically measure difficulty falling asleep, since they include awakenings after sleep onset. We therefore changed the principal exploratory Model B outcome to f_slp:fitbit_sleep_summary_rapids_sumdurationasleepmain:allday, the duration in minutes of the main sleep episode. This is a transparent outcome revision prompted by observed measurement limitations; the resulting association is interpreted as a main-sleep-duration relationship rather than an additional sleep-onset delay.

The primary exposure is f_screen:phone_screen_rapids_sumdurationunlock:evening. Only nonmissing exposure–outcome pairs are retained. The initial model restricts screen-unlock duration to 0–360 minutes and main-sleep duration to 180–840 minutes. To ensure that the entire evening exposure period precedes the target sleep onset, the main analysis uses recorded bedtimes of at least 1,440 minutes under RAPIDS's midnight-relative convention; earlier bedtimes are excluded and should be assessed in sensitivity analyses. Observations are organised chronologically within the anonymised participant. No

target values are imputed, and the within-person norm used for prediction is estimated only from each participant's training history. The 2018 dataset contains one cohort; cohort indicator variables are not identifiable and are omitted from its fitted model.

7.3 Exploratory Data Analysis of the Available GLOBEM Cohort

The 2018 GLOBEM feature panel consists of 14,260 participant-day rows across 155 participants. The evening screen-unlock duration is nonmissing for 9,960 rows (69.8%), whereas main-sleep duration is nonmissing for 7,476 rows (52.4%). Because the daily table includes dates without valid sensor measurements, the nominal panel size should not be confused with the number of usable participant-nights. Figure B4 plots the daily availability of the two measurements across stored dates.

Figure B1 shows the empirical distribution of the observed evening screen-unlock duration. This variable represents total unlocked-screen time during 18:00–24:00 rather than an identified social-media application or an observed pre-bed phone session (GLOBEM Dataset Authors, n.d.). Figure B2 shows the extreme zero inflation of the candidate sleep-onset measure: 7,443 out of 7,476 observed values (99.6%) equal zero. As this outcome fails the original quality threshold of less than 80% zeros, sleep-onset latency is not modelled as a continuous target. Fitbit's published sleep-data dictionary notes that its minutes-to-fall-asleep field will almost always be zero, reinforcing the measurement concern (Fitbit, 2024).

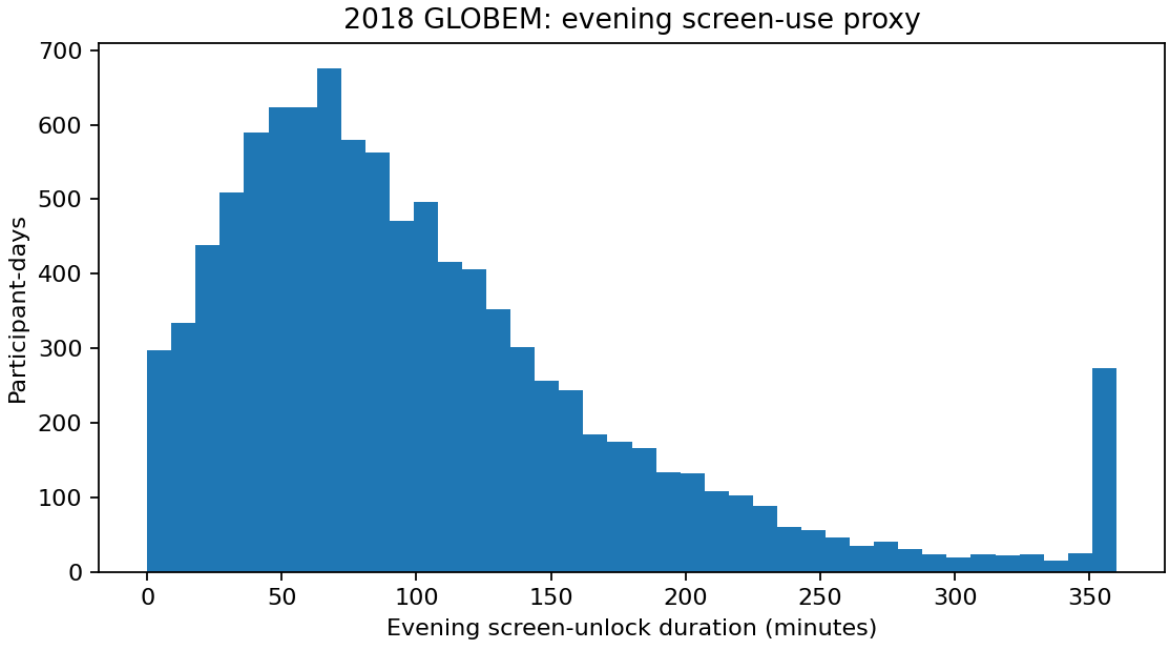


Figure B1. *Distribution of evening screen-unlock duration among observed 2018 participant-days.*

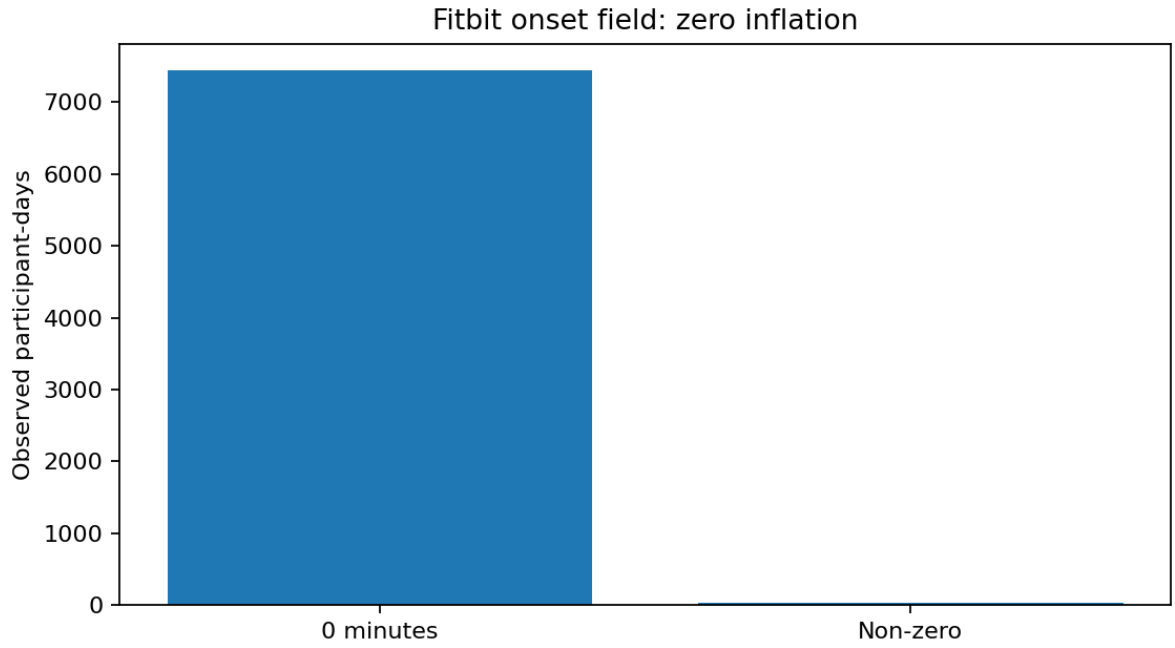


Figure B2. *Zero inflation in the Fitbit main-sleep sleep-onset field, observed values only.*

Figure B3 presents the distribution of Fitbit main-sleep duration. The observed median in the complete sleep feature is 429 minutes. Main-sleep duration is selected as an explicitly different exploratory outcome, not relabelled as sleep-onset delay. The conservative after-midnight eligibility restriction gives 4,718 participant-nights from 124 participants after minimum-history filtering. In this analysis sample, the median recorded main-sleep duration is 415 minutes and the median evening screen-unlock duration is approximately 89 minutes. Figure B5 uses aggregated bins of within-person centred data to visualise the descriptive relationship without releasing participant-level records. Figure B6 contrasts median sleep duration on Friday/Saturday nights with the remaining nights. These descriptive figures are not evidence of causality; potential confounding, platform imbalance and selection into the after-midnight sample remain important limitations.

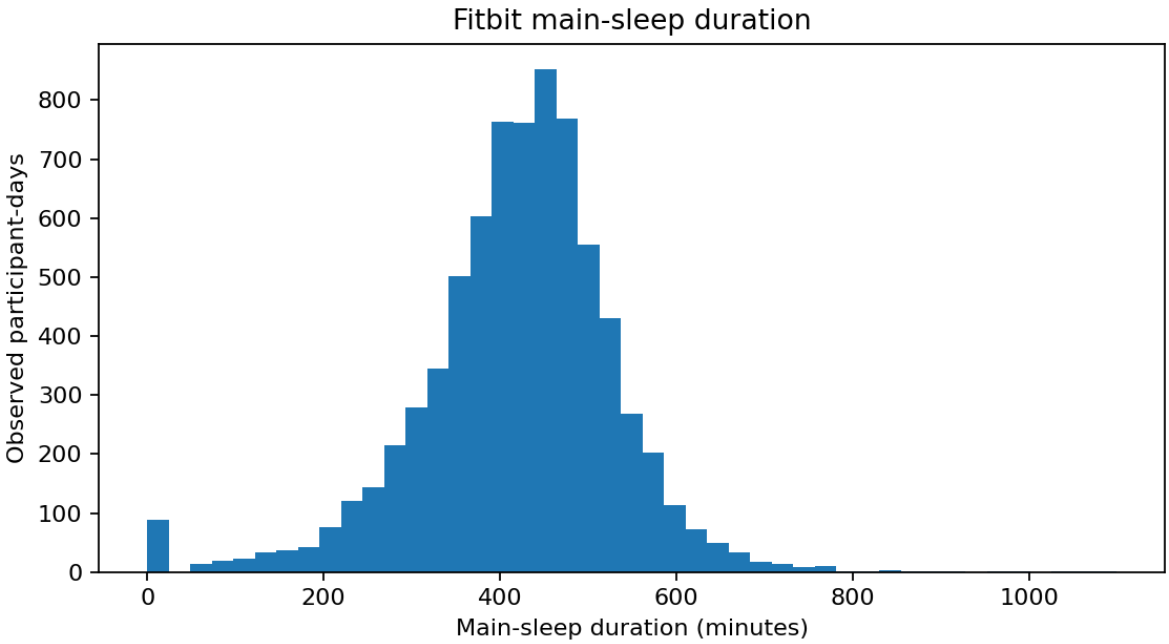


Figure B3. *Distribution of observed main-sleep duration, in minutes.*

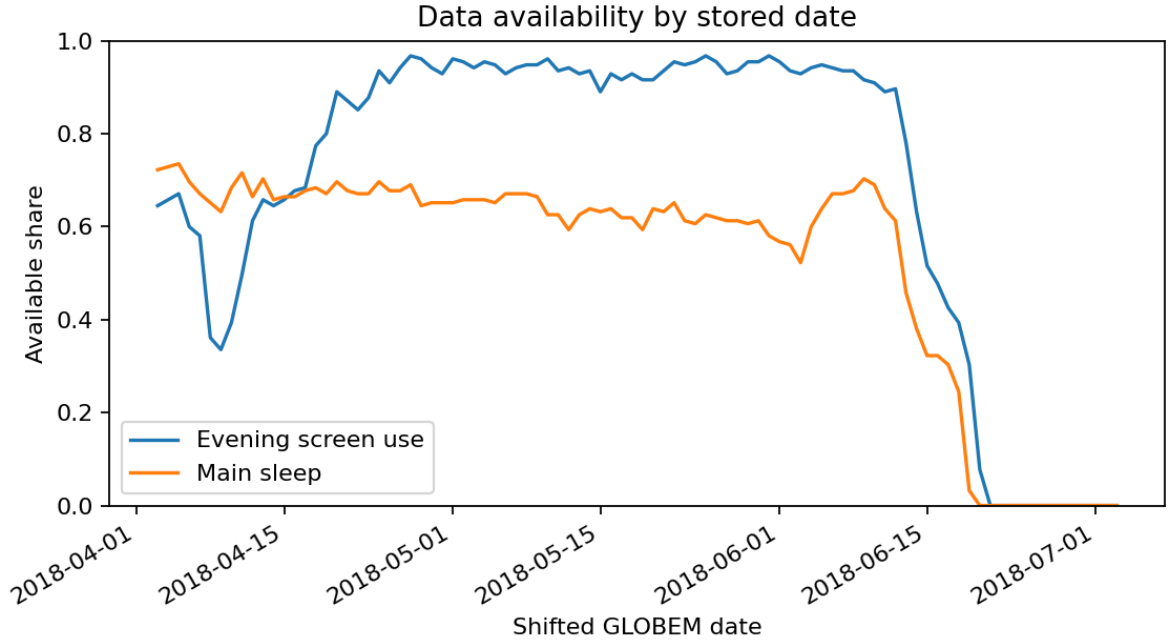


Figure B4. *Daily proportion of available screen and sleep features on privacy-shifted dates.*

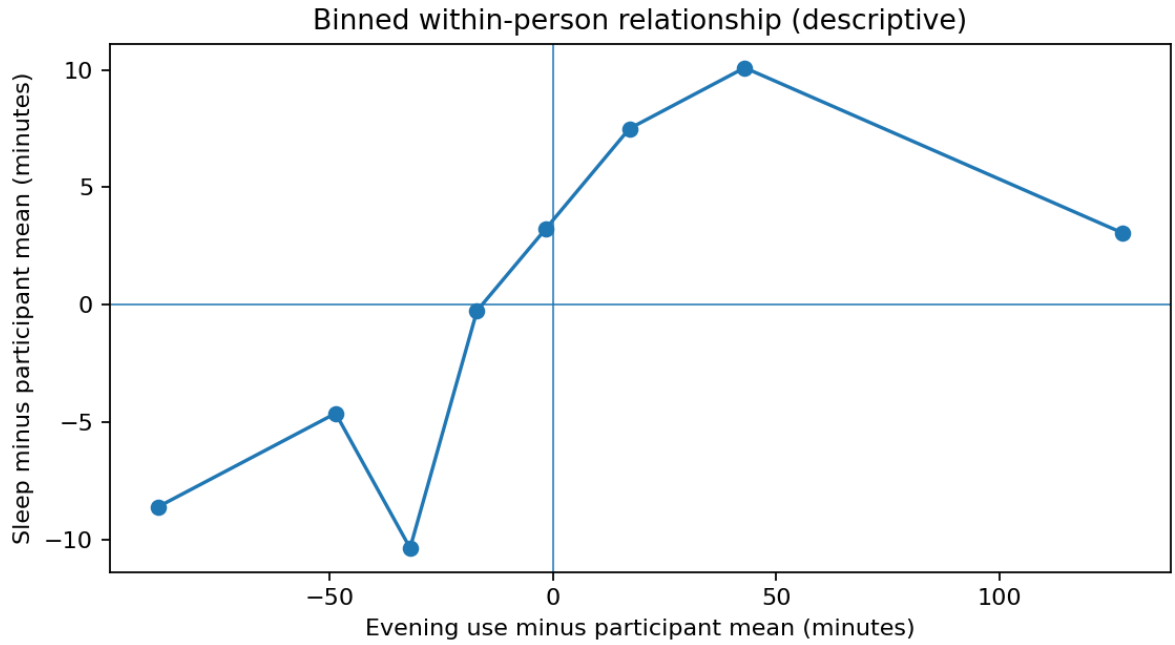


Figure B5. *Aggregated bin means for person-centred evening screen use and sleep duration.*

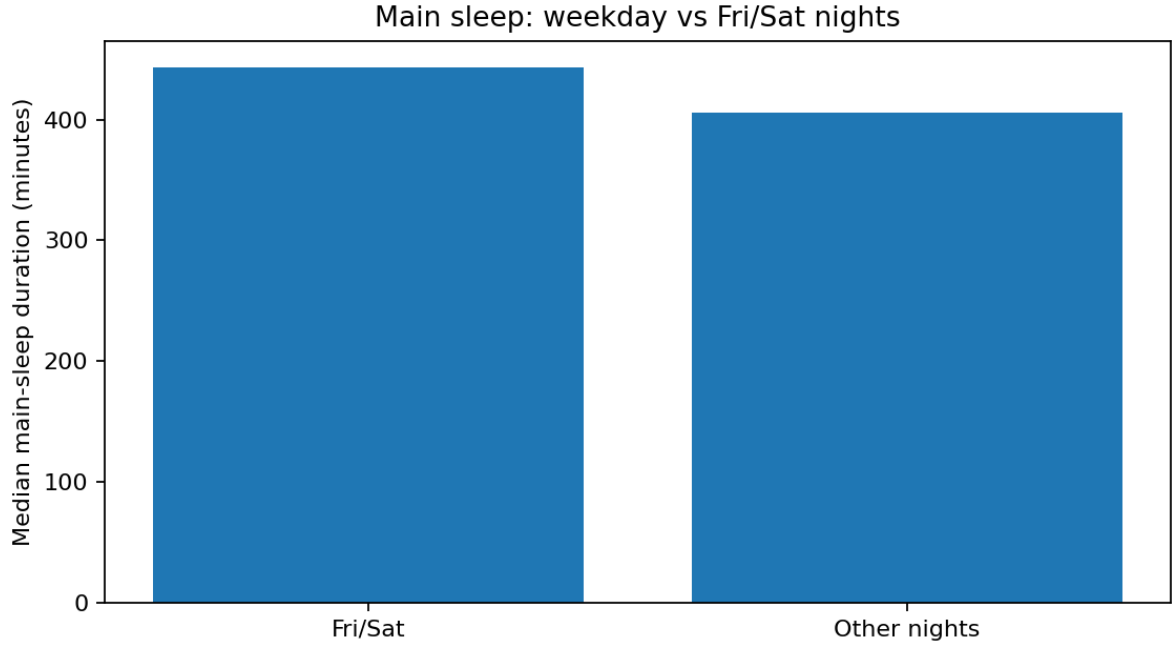


Figure B6. *Median main-sleep duration on Fri/Sat compared with other nights, eligibility-filtered sample.*

Supporting user survey (not model-training data). Alongside GLOBEM, we ran a short online survey of the target group (84 responses on 19–20 October 2026; convenience sample, 79 of 84 aged 18–25). Its self-reported responses are retained as evidence about the perceived utility and acceptability of the proposed nudge (Table 12); they are not merged into GLOBEM and are not presented as objectively measured sleep effects.

Table 12. *User survey results (n = 84) and their implications for Model B.*

Finding	Result	Implication for Model B
Phone use in bed	72 (86%) use the phone after getting into bed on at least 3 nights a week; short video is the most used content (61)	The exposure is common in the target group
Perceived sleep-onset delay	54 (64%) feel that falling asleep is delayed on those nights (32 clearly, 22 somewhat); 23 (27%) feel no delay	The effect is likely to differ between people, which supports person-specific terms
Self-estimate vs. screen-time record	Of 65 who checked their screen-time screen, 27 (42%) had estimated a lower range than the record showed and 7 a higher one	Self-reports understate use, so the app measures use passively and asks the user only about sleep
Reaction to an example warning	61 (73%) found it useful (22 would act on it, 39 would continue anyway); 17 (20%) would be bothered	Warnings need simple opt-out and mute settings

7.4 Feature Engineering — 2018 Implementation

Within-between design. A person's evening use on night i is split into their usual level (the person's mean) and the night's deviation from it. The deviation answers the app's question: what happens to this person on a night when they use the phone more than usual? The person mean absorbs stable differences between people (for example, heavy users who also sleep differently for other reasons), which would otherwise bias a single pooled slope.

Table 13. *Model B features (one row per participant-night).*

Feature	Definition	Rationale
use_within	(evening screen minutes − person's training-period mean) / 10	Within-person effect per extra 10 min of use; this coefficient drives the warning
use_between	(person's training-period mean − mean of the training participants' means) / 10	Separates stable between-person differences from the nightly effect
is_weekend	1 for Friday and Saturday nights	The next morning is usually free, so bedtime and sleep shift
cohort	Not used: the supplied data contain one cohort (2018)	Would separate study years; added only if the 2019–2021 files are obtained
platform (sensitivity)	iOS or Android	Screen use is measured differently on the two systems; with 145 iOS and 10 Android participants it is a sensitivity covariate only, and no Android-only estimate is reported
unlocks_within	Within-person deviation of the	Frequent checking versus one

(sensitivity)	evening unlock count	long session; not included in the first fitted model

Exposure window and reverse causation. The exposure is screen use in the evening segment (18:00–24:00), the closest GLOBEM proxy for use before bedtime. Night-segment use (00:00–06:00) is excluded from the primary model, because phone use after going to bed can be a result of not falling asleep as well as a cause. This is a limitation: many students go to bed after midnight, and the app's interface variable (use before the person's target bedtime, relative to their own norm) is narrower than a six-hour segment. The pilot data are used to re-check the link between the two definitions.

Leakage control. The analytic sample is defined using predetermined measurement-quality and minimum-history rules. For the chronological evaluation, the first 80% of each participant's eligible records constitute training data and the remainder form the test set. A participant's mean evening use is calculated from that person's training nights only and reused unchanged when transforming subsequent test nights. For group-held-out evaluation, unseen participants have no known personal norm and are assigned the mean training-participant norm; their personal random intercept is zero. This deliberately tests cold-start generalisation. Last night's outcome is not used as a feature.

7.5 Data Transformation — 2018 Implementation

- **10-minute units.** Screen-unlock use enters in units of ten minutes, so the within-person coefficient reads as sleep minutes per additional ten evening screen-unlock minutes relative to the training-period personal norm. Binary is_weekend is left unscaled.
- **No GLOBEM normalisation.** GLOBEM also provides normalised features (person median removed, divided by the 5–95% range). They are not used, because the phone needs a formula in plain minutes.
- **Centring** is the within-between split above; z-scoring is not needed for a linear model in these units.
- **Target scale.** The sleep-duration outcome remains in minutes and is not log-transformed in the reported exploratory fit; neither the target nor the predictor is z-scored. Adopting a log scale in future would change the coefficient interpretation and require back-transformation.
- **Export.** No Model B parameter file is shipped in the pilot. The fitted fixed effects and variance components are reported as aggregate results only (section 5.4 of Model Exploration).

Model Exploration

1. Model Selection

Because 6.3% of the snapshots are censored and the target is a duration, we treat Model A as a **survival (time-to-event) regression**, not as plain regression or classification. We compared models in order of increasing complexity:

Table 5. *Candidate models.*

Model	Strengths	Weaknesses
Baseline 1: personal median length	Trivial, interpretable: max(0, personal median length − t)	Ignores that the session has already survived to t; no interval
Baseline 2: personal remaining	Quartiles of R in the user's past sessions that reached t; uses survival information	Needs history (falls back to app, then global); no context features
Weibull AFT (lifelines) — ships	Handles censoring exactly; closed-form quantiles; 19 coefficients + 1 shape parameter fit in JSON; interpretable	Assumes a Weibull shape shared by all sessions; linear in features
Weibull AFT + personal offset — ships	Adapts to each user online with only two numbers per user; equals the population model for new users	One multiplier per user; cannot model user-specific feature effects
XGBoost AFT — benchmark	Non-linear, captures interactions	Hundreds of trees; hard to reproduce on the phone; weaker calibration

Weibull accelerated failure time (AFT). The model assumes R follows a Weibull distribution with scale $\lambda = \exp(\beta_0 + \beta^T x)$ and shape ρ. A coefficient β acts as a time multiplier: $\exp(\beta)$ is the factor by which the remaining time stretches for a one-unit (one standard deviation) increase of the feature. Any quantile has the closed form $r_q = \lambda \cdot (-\ln(1 - q))^{(1/\rho)}$, so the phone can compute the median and the 25–75% range with a few lines of Kotlin. The fitted shape $\rho = \exp(-0.138) = 0.87 < 1$ confirms the decreasing hazard seen in the EDA.

Personal offset (gamma frailty). Users differ systematically in how long they stay. Each user gets a multiplier θ on the Weibull hazard, $S(r) = \exp(-\theta (r / \lambda)^{\rho})$, with a Gamma(a, a) prior of mean 1. After the user's own sessions, the posterior mean is closed form: $\theta = (a + n_events) / (a + exposure)$, where every past session that reached 5 minutes adds $(R / \lambda)^{\rho}$ to the exposure and, if its end was seen, 1 to n_events. Censored sessions count through their exposure only, so censoring is handled exactly. The phone stores just these two numbers per user; with no history $\theta = 1$ and the prediction equals the population model. $\theta < 1$ means longer sessions than the model expects.

Models considered and not pursued. A Cox proportional hazards model was considered, but it needs a stored baseline hazard to produce quantiles and gives no advantage over the AFT for deployment. LSTM/GRU sequence models are future work. XGBoost AFT remains an offline benchmark and would only replace the Weibull model if it clearly won, which it does not (section 3).

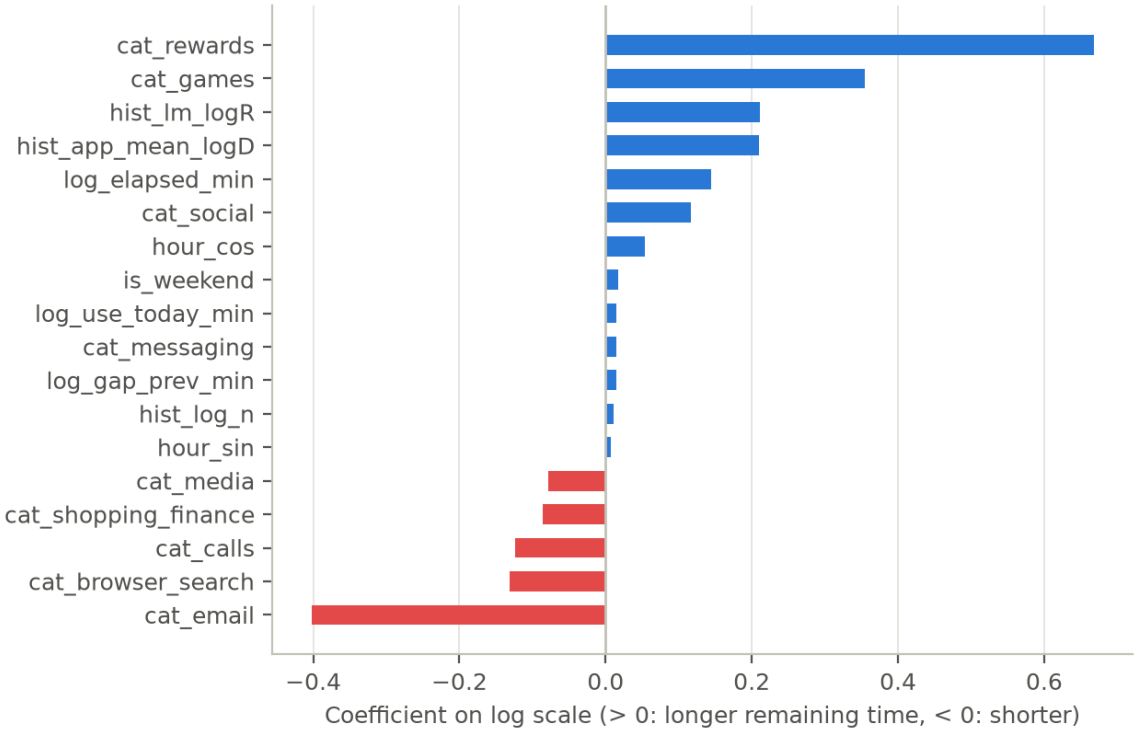


Figure 6. *Weibull AFT coefficients exported to params/model_a.json (numeric features standardised; categories relative to "utility").*

The coefficients are plausible (Figure 6): reward apps (×1.95) and games (×1.42) stretch remaining time, email (×0.67), browser/search and calls shorten it, and the two personal history features and elapsed time are the strongest numeric effects. Hour of day has a small direct effect once category and history are known.

2. Model Training

Validation design. Two splits are used, and all landmark snapshots of one session always stay in the same split:

- **Chronological 80/20 per user** (primary): the first 80% of each user's sessions by start time are used for training, the last 20% for testing. This mirrors deployment, where the model predicts a user's future from their past.
- **User-held-out 5-fold cross-validation** (GroupKFold on user_id): test users are never seen in training. This measures performance for a brand-new user and is reported as the mean over the five folds.

All preprocessing statistics (fill values, means, standard deviations, baseline lookup tables, frailty prior) are fitted inside each training split only.

Table 6. *Training configuration (seed 42).*

Model	Training details and hyperparameters
Weibull AFT	lifelines WeibullAFTFitter, L2 penalizer 0.001 (small, only for numerical stability); columns constant in the training split are dropped; maximum likelihood with right-censoring
Personal offset	Gamma prior shape a fitted by maximum marginal likelihood over users (bounded 1-D search on ln a). a = 36.5 on the chronological split (frailty variance 0.027), 29.4–35.1 across user folds; a = 32.0 for the exported model trained on all data. θ is updated online from sessions that ended before each prediction moment
XGBoost AFT	objective survival:aft, normal error distribution, scale 1.2, learning rate 0.05, max_depth 3, min_child_weight 20, subsample 0.8, colsample_bytree 0.8, up to 2,000 rounds with early stopping after 50 rounds on aft-nloglik, using a user-grouped 20% validation split of the training data
Baselines	Personal statistics need at least 5 sessions; otherwise they fall back to app level and then global level

The shipped parameters are produced by python -m ml.model_a.export, which refits the Weibull AFT and the frailty prior on all LSApp data and writes params/model_a.json together with synthetic parity cases that the Kotlin implementation must reproduce.

3. Model Evaluation

Why no confusion matrix or ROC curve. Model A predicts a continuous, partly censored duration, not a class, so classification diagnostics do not apply. We use the survival equivalents:

- **MAE and median absolute error** (minutes) of the predicted median on completed (uncensored) sessions.
- **Harrell's C-index**, which uses censored sessions too and measures whether sessions predicted to last longer actually do (0.5 = random).
- **Coverage and width of the 25–75% interval.** A calibrated interval contains 50% of completed sessions; width shows its sharpness.
- Results are reported at t = 5 (the main nudge point), t = 15 and t = 30, for the evening window at t = 5, and over all landmarks.

Table 7. *Results at t = 5 min (errors and width in minutes). Baseline 1 gives a point estimate, so it has no interval.*

Model	MAE	MedAE	C-index	Coverage	Width
Chronological 80/20 (n = 2,013)					
Personal median length (baseline 1)	4.10	1.98	0.507	–	–
Personal remaining (baseline 2)	3.29	1.44	0.562	0.458	3.98
Weibull AFT	3.22	1.50	0.566	**0.506**	4.11
Weibull AFT + personal offset	**3.18**	1.51	**0.580**	0.508	4.30
XGBoost AFT	3.21	**1.28**	0.571	0.451	3.24
User-held-out, 5-fold mean (n ≈ 1,867)					
Personal median length (baseline 1)	3.80	1.97	0.507	–	–
Personal remaining (baseline 2)	2.97	1.40	0.555	0.468	3.63
Weibull AFT	3.01	1.58	0.568	0.522	4.68
Weibull AFT + personal offset	**2.96**	1.54	**0.577**	0.522	4.68
XGBoost AFT	3.00	**1.30**	0.574	0.464	3.37

Table 8. *MAE in minutes by report group; best value per cell in bold (n = 180 / 148 at t = 15 and 36 / 28 at t = 30).*

Model (chronological / user-held-out)	t = 15	t = 30	t = 5 evening	All landmarks
Personal median length (baseline 1)	10.94 / 9.65	20.85 / 18.31	4.11 / 4.24	5.69 / 5.05
Personal remaining (baseline 2)	**7.48** / **7.57**	13.32 / **14.12**	3.23 / 3.31	4.28 / **3.94**
Weibull AFT	7.94 / 7.77	14.13 / 14.67	3.24 / 3.35	4.33 / 4.01
Weibull AFT + personal offset	7.50 / 7.68	**12.44** / 14.62	**3.20** / 3.29	**4.19** / 3.95
XGBoost AFT	8.33 / 7.88	15.46 / 15.06	3.21 / **3.28**	4.41 / 4.02

Table 8b. *MAE difference (minutes) of the Weibull AFT + personal offset against each alternative, with 95% user-cluster bootstrap intervals (2,000 resamples of users; user-held-out folds pooled). Negative favours the shipped model; bold intervals exclude zero.*

Shipped model vs	**Chronological, t = 5**	**Chronological, all**	**User-held-out, t = 5**	**User-held-out, all**
Personal remaining (baseline 2)	**−0.11** [−0.20, −0.05]	**−0.09** [−0.17, −0.03]	−0.01 [−0.04, +0.05]	+0.01 [−0.12, +0.20]
Weibull AFT	**−0.04** [−0.11, −0.01]	**−0.14** [−0.38, −0.02]	**−0.05** [−0.10, −0.01]	−0.06 [−0.16, +0.01]
XGBoost AFT	−0.04 [−0.07, +0.00]	**−0.23** [−0.48, −0.06]	−0.03 [−0.10, +0.01]	−0.07 [−0.17, +0.01]

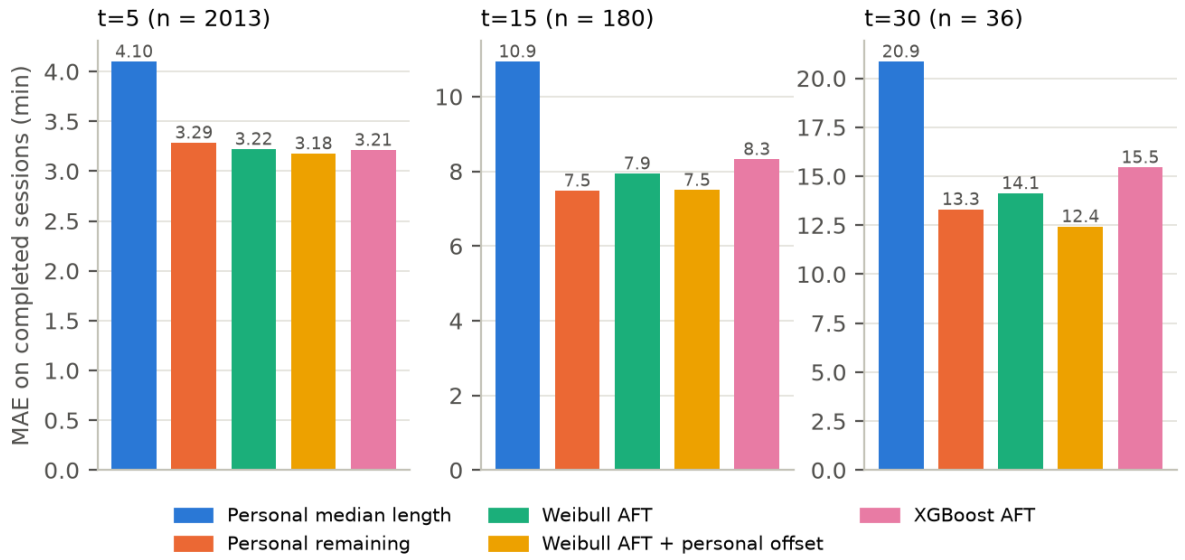


Figure 7. *MAE of the predicted median on the chronological split at t = 5, 15 and 30 minutes.*

Accuracy. All models beat the naive baseline 1 by a wide margin. The Weibull AFT with personal offset has the lowest MAE at t = 5 on both splits and over all landmarks on the chronological split, and the highest C-index at t = 5 and over all landmarks on both splits. Its advantage over the strong second baseline is **modest and not universal**: on the chronological split it is small but statistically reliable (−0.11 min at t = 5, 95% interval −0.20 to −0.05; Table 8b), whereas for new users (user-held-out) the two are indistinguishable (−0.01 min, interval −0.04 to +0.05), and baseline 2 is better at t = 15 and t = 30 on that split. The personal offset reliably improves on the plain Weibull AFT at t = 5 on both splits. XGBoost has the lowest median absolute error but no better MAE, and its advantage disappears at longer elapsed times. Results at t = 30 rest on only 36 (chronological) and about 28 (user-held-out) test sessions and should be read as indicative only.

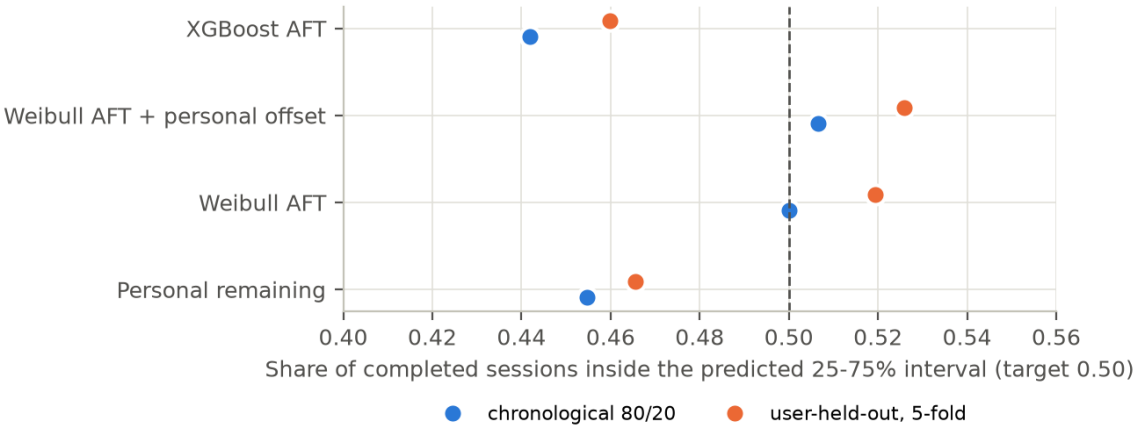


Figure 8. *Coverage of the predicted 25–75% interval over all landmarks (target 0.50).*

Calibration. Over all landmarks, the Weibull intervals contain 0.50–0.53 of completed sessions against a target of 0.50, while baseline 2 and XGBoost are too narrow (0.44–0.47), that is, over-confident (Figure 8). This coverage is computed on completed sessions only; censored sessions, which tend to be long, are excluded, so it does not by itself establish calibration under censoring. Calibration of the median against Kaplan-Meier estimates, which do use censored sessions (Figure 9), is close to the diagonal for seven of eight prediction bins. In the top bin the model predicts about 5.5–6.4 min where the observed median is about 10 min: the longest sessions are still under-predicted, and the personal offset reduces but does not remove this gap.

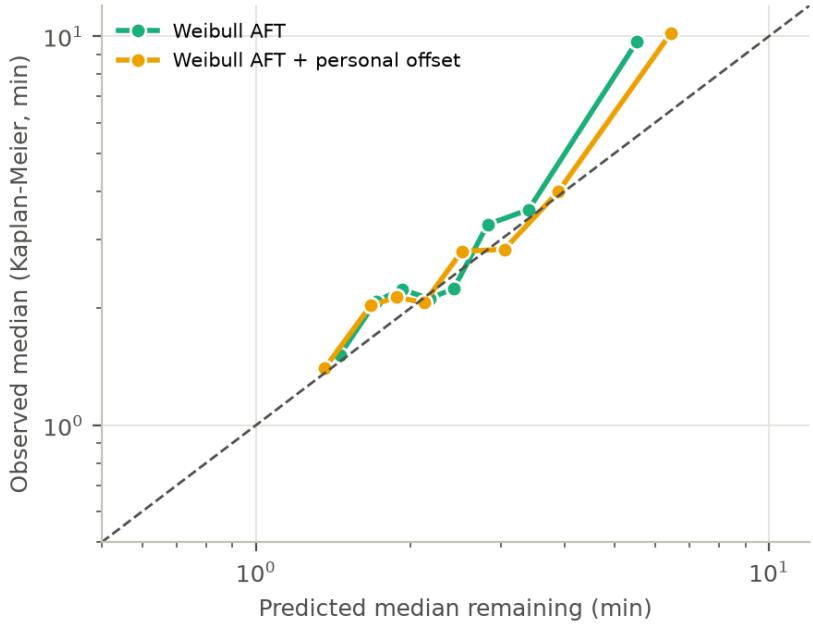


Figure 9. *Predicted versus observed (Kaplan-Meier) median remaining time in eight bins of the prediction, chronological test split.*

Personalisation. Among the 186 test users with history, the final personal multiplier θ ranges from 0.82 to 1.16 (5th–95th percentile; median 1.01, Figure 10). Most users are close to the population model, and a minority have clearly longer or shorter sessions. For them the offset corrects predictions without any per-user retraining.

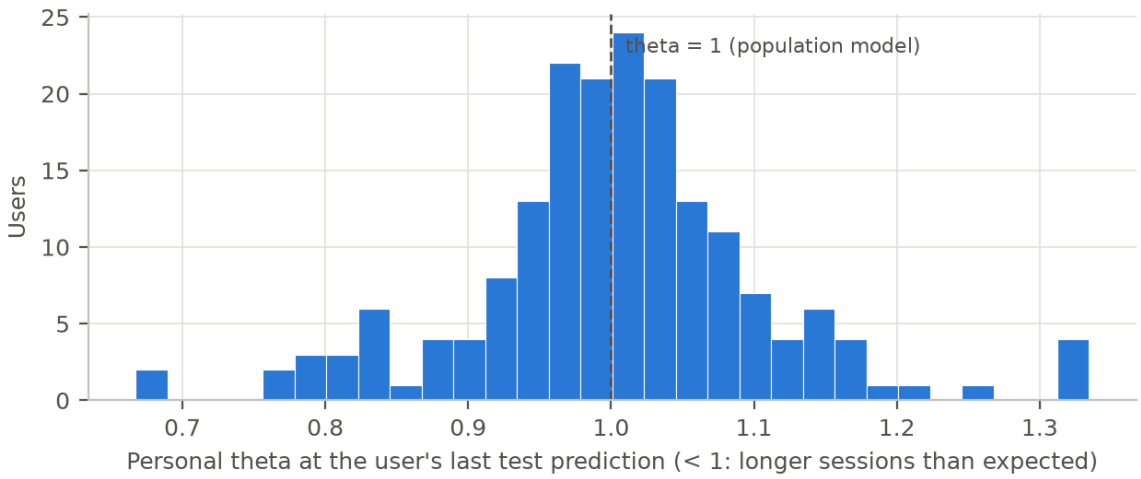


Figure 10. *Distribution of the personal multiplier θ at each user's last test prediction.*

Sensitivity: without reward apps. Removing reward apps leaves 11,607 snapshots (72 at t = 30). Errors drop for every model, for example from 3.18 to 2.96 min at t = 5 (chronological) for the Weibull AFT with personal offset. It remains the best model at t = 5 on both splits (2.96 / 2.64 min) and over all landmarks, and without reward apps its advantage over baseline 2 is statistically reliable on both splits (user-held-out, t = 5: −0.03 min, interval −0.06 to −0.00). Whether to exclude or flag reward apps in the shipped model is still an open team decision.

Conclusion. The Weibull AFT with a gamma-frailty personal offset is selected as an interpretable, compact candidate for deployment: it has the best MAE and discrimination at the main five-minute landmark, the best-calibrated central interval among the compared models (which the nudge text relies on), and needs only a JSON file plus two numbers per user on the phone. Its advantage over the strong personal-remaining-time baseline is modest and not universal: it is reliable for returning users but not demonstrated for new users, and it does not hold in every subgroup. The very longest sessions remain under-predicted because LSApp contains few sessions over 30 minutes; this should improve as the pilot collects the users' own data.

4. Code Implementation

The full code is in the project repository (ml/model_a/). Run order from the repository root: python -m ml.model_a.train (evaluation tables), python -m ml.model_a.export (shipped parameters and parity cases) and python -m ml.model_a.report_figures (figures). The key sections are shown below.

Session reconstruction and cleaning (sessions.py):

```
def drop_consecutive_duplicates(events):
    # Rule 1: drop a row only if it is identical to the row directly before it
    same_as_prev = (events[COLUMNS] == events[COLUMNS].shift()).all(axis=1)
    return events[~same_as_prev].reset_index(drop=True)

def build_sessions(events):
    # Rules 2-3: pair Opened/Closed per (session_id, app_name); unmatched
    # sessions are kept as right-censored. Rule 4: merge re-entries within 30 s.
    s = merge_reentries(pair_events(drop_consecutive_duplicates(events)))
    s["duration_s"] = (s.end - s.start).dt.total_seconds()
    s["duration_min"] = s.duration_s / 60
    s["event"] = (~s.censored).astype(int)                # 1 = end observed
    s["local_start"] = s.start + pd.Timedelta(hours=-5)     # Rule 5
    return s.sort_values(["user_id", "start"]).reset_index(drop=True)
```

Landmark snapshots and leakage-free history features (features.py):

```
def add_session_history(s):
    ...
    # Completed sessions only (a censored duration is a lower bound); shift() so a
    # session never sees its own duration. Same-app sessions never overlap.
    s["hist_app_mean_logD"] = s.logD.where(s.event == 1).groupby(
        [s.user_id, s.app_name]).transform(lambda x: x.shift().expanding().mean())
    s["use_today_min"] = s.groupby([s.user_id, day]).duration_min.transform(
        lambda x: x.shift(fill_value=0).cumsum())
    s["use_today_min"] -= _running_overlap_min(s, day)  # minutes after this start

def past_landmark_logR(L):
    # Only completed sessions that ENDED before the prediction moment start + t
    moment = (L.start + pd.to_timedelta(L.t, unit="m")).to_numpy()
    for idx in L.groupby(["user_id", "t"]).indices.values():
        done = L.iloc[idx].query("event == 1").sort_values("end")
        k = np.searchsorted(done.end.to_numpy(), moment[idx], side="left")
        csum = np.concatenate([[0], np.log(done.R.to_numpy()).cumsum()])
        out[idx] = np.where(k > 0, csum[k] / np.maximum(k, 1), np.nan)

for t in [5, 10, 15, 20, 30]:                    # landmark snapshots
    alive = s[s.duration_min > t].copy()          # still open at t minutes
    alive["t"] = t
    alive["R"] = alive.duration_min - t           # target: remaining minutes
```

Weibull AFT fit and closed-form quantiles (train.py); the quantile formula is the one Kotlin uses:

```
def fit_weibull(X, d):
    X = X.loc[:, X.nunique() > 1]           # drop columns constant in this split
    return WeibullAFTFitter(penalizer=0.001).fit(
        X.assign(R=d.R.values, event=d.event.values), "R", "event")

def weibull_quantiles(lambda_coef, rho_log, X, qs=(0.25, 0.5, 0.75)):
    """r_q = lambda * (-ln(1 - q))^(1/rho); the same formula Kotlin uses."""
    lam = np.exp(lambda_coef["Intercept"] + sum(
        X[c].to_numpy() * b for c, b in lambda_coef.items() if c != "Intercept"))
    rho = np.exp(rho_log)
    return np.column_stack([lam * (-np.log(1 - q)) ** (1 / rho) for q in qs])
```

Personal offset (personal.py):

```
def exposure(R, lam, rho):
    return (R / lam) ** rho          # censored sessions count through exposure only

def fit_prior_shape(u):
    # Gamma(a, a) prior; a by maximum marginal likelihood over users
    per_user = u.groupby("user_id")[["d", "e"]].sum()
    D, E = per_user.d.to_numpy(), per_user.e.to_numpy()
    def nll(log_a):
        a = np.exp(log_a)
        return -np.sum(gammaln(a + D) - gammaln(a) + a * np.log(a) - (a + D) * np.log(a + E))
    return float(np.exp(minimize_scalar(nll, bounds=(-5, 10), method="bounded").x))

def theta(a, n_events, expo):
    return (a + n_events) / (a + expo)   # = 1 for a new user

def quantiles(lam, rho, th, qs):
    return np.column_stack([lam * (-np.log(1 - q) / th) ** (1 / rho) for q in qs])
```

Evaluation split and metrics (evaluate.py):

```
def chronological_split(s, train_frac=0.8):
    # True for the first 80% of each user's sessions by start time
    rank = s.groupby("user_id").start.rank(method="first")
    n = s.groupby("user_id").start.transform("size")
    return rank <= train_frac * n

def metrics(pred_q, te):
    R, E = te.R.to_numpy(), te.event.to_numpy()
    lo, med, hi = pred_q[:, 0], pred_q[:, 1], pred_q[:, 2]
    done = E == 1                            # errors on completed sessions
    err = np.abs(med[done] - R[done])
    return dict(mae=err.mean(), medae=np.median(err),
                c_index=concordance_index(R, med, E),   # uses censored rows too
                coverage=np.mean((R[done] >= lo[done]) & (R[done] <= hi[done])),
                width=np.mean(hi - lo))
```

5. Model B: Model Exploration (2018 GLOBEM Pilot)

Three requirements drive the choice of model for Model B: it must separate a person's nightly effect from stable differences between people, give an uncertainty range, and update for each user on the phone with a few arithmetic operations. As for Model A, every candidate must beat a simple personal baseline before it is used. The previously described random-intercept linear mixed model has now been fitted to a temporally restricted 2018 analysis sample. The model estimates a participant-specific intercept and one shared within-person association between evening unlock duration and main-sleep duration. This is an exploratory evaluation of a revised sleep-duration outcome, not a validated sleep-onset-delay predictor. The random-slope and gradient-boosting alternatives remain candidates for future comparison and are not claimed to have been trained in this analysis.

5.1 Model Selection and Interpretation

Table 14. *Candidate models for Model B.*

Model	Strengths	Weaknesses
Baseline: personal mean	Mean target over the person's earlier nights (population mean for a new user); trivial and robust	Ignores how much the phone was used that night, so it cannot drive a warning
Linear mixed model, random intercept (primary)	Separates within- and between-person effects; closed-form prediction and personal update; a handful of numbers in JSON	Assumes the same nightly effect for everyone and a linear dose-response
Mixed model with a random slope on use_within	Lets sensitivity to phone use differ by person, as the survey suggests	Needs many nights per person; may not converge; more parameters on the phone
Gradient boosted trees (benchmark)	Non-linear; captures interactions	No clean within-person effect; hard to personalise and to reproduce on the phone

Linear mixed model. For participant j on night i, the model is $\text{sleep_min}_{ij} = \beta_0 + \beta_W \cdot \text{use_within}_{ij} + \beta_B \cdot \text{use_between}_j + \beta_{wk} \cdot \text{is_weekend}_{ij} + u_j + \varepsilon_{ij}$, with $u_j \sim N(0, \tau^2)$ and $\varepsilon_{ij} \sim N(0, \sigma^2)$. There is no cohort term, because the data contain one cohort. The intercept and the participant-specific offset contribute to the absolute prediction of main-sleep duration. However, comparing two otherwise identical hypothetical evenings for the same person gives Δ predicted sleep_min = $\beta_W \times \Delta$ evening_use_min / 10; the fixed participant offset cancels and must not be added to that contrast. Because the data are observational, this estimated contrast is an association and is not a predicted causal effect of changing phone use. A positive coefficient is not evidence that extending phone use improves sleep.

Personal offset. If an outcome matching the fitted Fitbit target is later measured on the phone, each morning the phone would compute the user's residuals e_i (observed target minus the population prediction) and update $\hat{u} = \Sigma e_i / (n + \sigma^2/\tau^2)$, with posterior variance $1 / (n/\sigma^2 + 1/\tau^2)$, as set out in the concept note. With no nights the offset is 0 and the prediction equals the population model; after many nights it approaches the user's mean residual. It plays the same role as Model A's gamma frailty, with one difference in use: it moves the user's predicted level, not the size of the extra-use effect Δ.

Models considered and not pursued. GLOBEM is observational, so β_W is reported as a within-person association; a causal claim would need an experiment. Sequence models over daily data are left out for the same reason as in Model A: they cannot be reproduced cheaply on the phone.

5.2 Model Training

The analysis uses 4,718 eligible participant-nights from 124 people after missingness, duration, bedtime and minimum-history restrictions. For the primary deployment-like validation, observations are sorted by date within each participant, with the earliest 80% assigned to training and the remaining records reserved for testing. This yields 3,725 training observations and 993 test observations. All per-person exposure means are computed using training records only. A random-intercept model was fitted by restricted maximum likelihood using statsmodels MixedLM with the Powell optimiser; the fit converged. The seven-record threshold is a cohort inclusion criterion based on total eligible nights, not a guarantee of seven earlier training nights for every test prediction; two included participants contribute fewer than seven training records. A stricter seven-training-night warm-start analysis remains a sensitivity check.

An additional participant-held-out five-fold GroupKFold evaluation assesses prediction for users not represented in training. Group identifiers are unique participant IDs. In each fold, personal exposure norms for unseen individuals are replaced by the mean of the training-participant norms and no unseen-user random intercept is estimated. The original leave-one-cohort-out plan cannot be conducted with the current single-cohort extract and is reserved for a future multi-cohort analysis. The current evaluation uses one 2018 cohort and consequently contains no cohort fixed effects.

Table 15. *Model B training configuration (seed 42).*

Item	Setting
Library	statsmodels MixedLM; fixed effects use_within, use_between and is_weekend, with no cohort term
Estimation	REML for the reported coefficients; maximum likelihood when sets of fixed effects are compared (likelihood-ratio test, AIC)
Random slope	Not fitted in the 2018 pilot; remains a candidate for future comparison
Preprocessing	Person means are fitted on training nights only; eligibility filters (missingness, plausible ranges, bedtime ≥ 1,440 min, at least seven eligible nights) are applied before the split
Optimiser	Powell; the REML fit converged

5.3 Model Evaluation — Preliminary 2018 Results

- **MAE and median absolute error** (minutes) of the predicted target on test nights, compared with the personal-mean baseline.
- **Coverage and width of the 25–75% prediction interval**, reported as a calibration diagnostic; the Model A and Model B ranges are not combined into one interval.
- **The within-person effect β_W** with its 95% confidence interval, and the intraclass correlation.

At the chronological held-out test point, the personal-mean baseline yields MAE = 66.29 minutes and median absolute error = 51.50 minutes. The fitted random-intercept mixed model yields MAE = 66.08 minutes and median absolute error = 52.15 minutes (n = 993 test nights; Table 16). The 0.21-minute MAE reduction is small. A participant-cluster bootstrap of the paired held-out absolute-error difference

gives an approximate 95% interval of −0.94 to +1.34 minutes, which includes zero; the primary comparison therefore does not establish a reliable predictive improvement over the personal-mean baseline.

In participant-held-out five-fold validation, the population-mean cold-start baseline has mean MAE = 69.41 minutes, compared with 68.24 minutes for the mixed model using fixed-effects predictions for unseen people. The fitted within-person coefficient is β_W = +0.689 sleep minutes per additional ten evening screen-unlock minutes, with an approximate model-based 95% confidence interval from +0.327 to +1.052 minutes. The fitted intraclass correlation is approximately 0.226, indicating that some outcome variation is attributable to stable between-person differences. The observed sign is positive, not a finding that higher evening use leads to shorter sleep. Unmeasured confounding, outcome selection, temporal-window restrictions and observational design prevent causal interpretation.

A plug-in normal 25–75% interval using the estimated residual standard deviation contains approximately 48.4% of observed held-out sleep outcomes. This is a preliminary calibration diagnostic, not a fully uncertainty-adjusted deployed predictive interval, because parameter and personal-effect uncertainties have not been fully propagated. As Model A and Model B make different predictions using different data sources, their respective quantiles must not be directly added or presented as a calibrated combined 25–75% range.

Table 16. *Model B validation results (2018 pilot, main-sleep duration).*

Evaluation / metric	**Personal-mean baseline**	**Mixed model**
Chronological split MAE, n = 993	66.29 min	66.08 min
Chronological split MedAE	51.50 min	52.15 min
Participant-held-out 5-fold mean MAE	69.41 min	68.24 min
Within-person slope β_W, 95% CI	Not applicable	+0.689 [+0.327, +1.052] sleep min per 10 use min
Primary-split MAE improvement, bootstrap 95% CI	—	−0.94 to +1.34 min
Approximate Q25–Q75 plug-in coverage	Not estimated	48.4%

Deployment decision. The originally planned numerical sleep-onset delay estimate is disabled: the sleep-onset target fails measurement validity and the revised main-sleep-duration mixed model does not show a robust primary-split error improvement. Model A may still issue a session-duration or user-configured bedtime-overrun warning. Any claim about a quantified sleep effect requires a validated outcome, independent predictive improvement and appropriate external testing.

5.4 Deployment Implications

Model B is not yet approved for a numerical sleep-onset or sleep-loss claim in the Android interface. The current GLOBEM result uses Fitbit main-sleep duration, whereas the proposed morning diary directly measures self-reported sleep timing and perceived sleep-onset latency. Those labels are not interchangeable: the same personal offset cannot be updated with a different target unless the measurement agreement is validated. The planned diary remains useful for feasibility testing and prospective data collection, but no sleep-effect coefficient should be shipped until the chosen deployed

measurement matches the fitted outcome and a separate held-out evaluation confirms useful performance. The application can continue to provide Model A remaining-session estimates and configured-bedtime notifications without attributing a causal sleep delay to phone use. Three parts of the app remain in the pilot:

- **Morning sleep diary.** One notification each morning opens a short diary adapted from the Consensus Sleep Diary (Carney et al., 2012): time in bed, time of trying to sleep, minutes to fall asleep, number of awakenings, final wake time and sleep quality (1–5). Its answers are collected for feasibility testing and future validation; they do not update a Model B offset, because they do not measure the fitted Fitbit outcome; Health Connect sleep sessions are read as supporting data when a wearable is available.
- **Exposure on the phone.** The previous evening's use before the target bedtime comes from UsageStatsManager (shared with Model A) and is expressed relative to the user's own norm.
- **Settings.** Target bedtime and wake time, diary reminder time, warnings and diary on or off, mute for tonight, and delete all data. The 20% of survey respondents who would be bothered by a warning make these controls a requirement.

Model B therefore stays switched off in the pilot: the diary collects labels for later validation, and no sleep number is shown.

5.5 Code Implementation

A reproducible 2018 exploratory analysis script accompanies this report. It imports only the designated evening-screen and main-sleep RAPIDS features; performs eligibility and overnight-timing filters; computes leakage-safe within-person features; fits the restricted-maximum-likelihood mixed model; evaluates chronological and participant-held-out splits; and exports only aggregate metrics and figures. It does not export participant-level records. Model B numerical deployment remains future work rather than an implemented feature of the pilot. Its core functions are shown below.

```
def add_exposure_features(n, train_mask):
   # n: one row per participant-night (pid, date, evening_use_min, target)
   # person norms come from training nights only, so a test night never sees itself
   norm = n[train_mask].groupby("pid").evening_use_min.mean()
   grand = norm.mean()
   n["person_norm"] = n.pid.map(norm).fillna(grand)          # new person -> population mean
   n["use_within"] = (n.evening_use_min - n.person_norm) / 10   # 10-min units
   n["use_between"] = (n.person_norm - grand) / 10
   n["is_weekend"] = n.date.dt.dayofweek.isin([4, 5]).astype(int)   # Fri/Sat night
   return n

def fit_model_b(train):
   # random intercept per person, REML with the Powell optimiser
   m = smf.mixedlm("target ~ use_within + use_between + is_weekend",
                   train, groups=train["pid"])
   return m.fit(reml=True, method="powell")

def personal_offset(residuals, sigma2, tau2):
   # normal-normal update; equals 0 (population model) for a new user
   n = len(residuals)
   u_hat = np.sum(residuals) / (n + sigma2 / tau2)
   u_var = 1 / (n / sigma2 + 1 / tau2)
   return u_hat, u_var

def extra_use_change(beta_w, se_w, d_min):
   # change versus the user's own normal night; the personal offset cancels out
   delta = beta_w * d_min / 10
   half = 1.96 * se_w * d_min / 10
   return delta, (delta - half, delta + half)
```

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